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KICP Fellowship Committee
University of Chicago

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Dear Members of the KICP Fellowship Committee,

I am writing to express my strong interest in the KICP Fellowship at University of Chicago. I am a PhD Candidate at the University of Zurich in the research group of Prof. Marcelle Soares-Santos, expecting to finish my PhD no later than August 2027. During my PhD, I have cultivated a unique, independent research program focused on constraining cosmology using electromagnetic and gravitational wave data, developing extensive expertise in astronomical instrumentation, gravitational-wave cosmology, and multi-messenger astrophysics. I believe that my previous work, my expertise in these research areas, and my experience and continued engagement in outreach and science education will help to develop a deeper understanding of dark energy using electromagnetic and gravitational wave data at the University of Chicago.

Shortly after beginning my studies in Zurich, I moved to La Serena, Chile to support the Vera C. Rubin Observatory through the final stages of construction and early science operations. While in Chile, I led the commissioning of the LSST-Camera focal plane array, including routine operation, intervention, and optimization of the LSST-Camera CCDs. I am eager to apply my instrumentation expertise to develop future spectroscopic instruments, focusing on sensor characterization for the Dark Energy Spectroscopic Instrument Run II and the Spectroscopic Stage-V experiment.

In parallel to commissioning the LSST-Camera, I coordinate several observational programs, with a focus on constraining cosmological parameters using observations from optical telescopes and gravitational wave detectors. This includes leading Target-of-Opportunity searches with Vera C. Rubin Observatory and the Dark Energy Camera using the LSST Science pipelines to process obtained data, and augmenting photometric searches using spectroscopy from the Prime Focus Spectrograph. Transitioning my expertise to Chicago will sustain the aforementioned observational programs under a common processing framework, helping to characterize electromagnetic counterparts of gravitational wave events and leverage an emerging cosmic probe to gain a deeper understanding of dark energy.

Throughout my research career, I have prioritized public engagement and outreach as a core complement to my scientific work. During the 2024 solar eclipse, I led a group of high school students from my hometown in guided lectures on solar physics, eclipse safety, and observation of the eclipse from the path of totality using a small solar telescope. In 2025, I provided interviews to media outlets covering the first observations of Vera C. Rubin Observatory, including video features with the BBC, and written interviews with the Wall Street Journal and Science News. Educational outreach and public engagement has reinforced that translating complex physics concepts for non-specialist audiences not only broadens the impact of my research, but also sharpens my own understanding of core physics concepts. I am committed to extending these broad impacts by serving as a mentor for University of Chicago physics PhD students, delivering public-facing talks at the Adler Planetarium and Astronomy and Astronomy on Tap Chicago, and serving as a mentor for the Space Explorers Program for Chicagoland high school students.

Enclosed is my CV, research statement, and publication list. Thank you for your consideration. I welcome the opportunity to discuss how my expertise could contribute to the scientific mission of University of Chicago.

All the Best,



Sean Patrick MacBride